

Chapters	Topics	Book to follow
Introduction to Probability	Random experiment, Sample space, Event space, Mutually exclusive events, collectively exhaustive events, disjoint events, Definition of probability - classical, relative frequency, subjective, probability axioms, discrete and continuous set, joint probability, conditional probability, product and sum rules of probability, chain rule, independent events, Bayes theorem, conditional product and sum rule, conditional Bayes theorem	Sheldon Ross/Goodman and Yates/Saeed Ghahramani
Random Variable	Definition, types - discrete, continuous, mixed, probability distribution function, probability models, probability mass function(PMF), cumulative distribution function(CDF), complementary cumulative distribution function(CCDF), continuous random variable, probability density function, cumulative distribution function, expected value, derived random variable, moment, nth moment, nth central moment, variance	Sheldon Ross, Saeed Ghahramani
Discrete Random variable	Bernoulli RV, Bernoulli Trials, Geometric distribution, binomial distribution, hyper geometric distribution, poisson distribution, pascal RV	Sheldon Ross, Saeed Ghahramani
Continuous Random Variable	Uniform Continuous RV, Exponential CRV, Gamma RV, Beta RV, Erlang distribution, Normal/gaussian RV, Standard normal RV, Table for probabilities, standardization of normal RV, Linear combination of normally distributed RVs, Lognormal distribution, Triangular distribution, Laplace distribution	Sheldon Ross, Saeed Ghahramani
Bivariate Random Variable	Discrete, continuous, mixed, Joint probability mass function(JPDF), joint cumulative distribution function(JCDF), marginal probability mass function,	Goodman and Yates, Saeed Ghahramani

	marginal cumulative distribution function, Expected value and variance from JPMF, Bivariate CRV, JCDF, JPDF, Expected value of function of 2 RVs, Covariance, Correlation, Correlation coefficient, Orthogonal RVs, Uncorrelated RVs	
Multivariate Random Variables	Multivariate JPMF(MJPMF), Multivariate JPDF(MJPDF), MJCDF, Marginal J/PMFs, J/PDFs, CDFs, Multivariate Discrete RVs, Multivariate hypergeometric distribution, Multinomial distributions, Multivariate CRVs, Marginal density functions	Goodman and Yates, Saeed Ghahramani
Conditional Distributions	Conditional probability, Product rule, sum rule, Bayes Theorem, Independence revisit, distribution conditioned on event, conditioned on RV, Conditional expectation, law of total expectation, mean, variance, continuous conditional distribution, expectation, independence of discrete random variables, CRVs	Goodman and Yates, Sheldon and Ross
Random Vectors and Gaussian RVs	Random vector, probability model, PDF, CDF, PMF, Joint probability models of 2 Random vectors, independent random vectors, expected value vector, correlation matrix, covariance, cross correlation, cross covariance,	Goodman and Yates
Sampling Statistics	Descriptive and Inferential stats, population, sample, measures of central tendency, mean, sample median, sample mode, measures of variability(dispersion), mean deviation, variance, standard deviation, statistics, sample mean, central limit theorem, approximation of binomial RVs, approximate distribution of sample mean, sample variance, sampling distribution from normal population, chi-square distribution, joint distribution, t-Distribution,	Sheldon Ross, Goodman and Yates

Parameter Estimations	Maximum Likelihood Estimators, Confidence interval(CI), one sided CI, CI with a certain level, CI for unknown Variance, upper and lower CI, CI for Variance of Normal Population, CI for the Differences in Means of 2 Normal Populations, Approximate Confidence Interval for the Mean of a Bernoulli Random Variable, Confidence Interval of the Mean of the Exponential Distribution, Evaluating a Point Estimator, The Bayes Estimator	Sheldon Ross, Goodman and Yates
Hypothesis Testing	Null Hypothesis, Alternative Hypothesis, Two-sided hypothesis testing, One-sided hypothesis testing, Steps to Test Hypothesis, Test for the Mean of Normal Population, Test for the Mean of Normal Population, p-value of the test, Probability of Type II error, Case of unknown Variance: the t-test, Testing the equality of means of two normal populations, The paired t-test, Hypothesis tests concerning the variance	Sheldon Ross
Multi-variate Gaussian distributions	Joint PDF, random vector, expectation vector, covariance matrix	Goodman and Yates